

The limit;

$$L = \lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} \frac{e^x - 1}{1 + n^4 x^4} dx$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \int_0^{\frac{1}{n}} \frac{e^x - 1}{x^4 + \left(\frac{1}{n}\right)^4} dx$$

Redefining $\frac{1}{n}$ as t ;

$$L = \lim_{t \rightarrow 0} t^2 \cdot \int_0^t \frac{e^x - 1}{x^4 + t^4} dx$$

$$= \lim_{t \rightarrow 0} \frac{1}{t^2} \cdot \int_0^t \frac{e^x - 1}{1 + \left(\frac{x}{t}\right)^4} dx$$

However; recall that:

$$e^x - 1 = \sum_{m=1}^{\infty} \frac{x^m}{m!}$$

$$\frac{1}{1 + \left(\frac{x}{t}\right)^4} = \sum_{n=0}^{\infty} \left(\frac{x}{t}\right)^{4n} (-1)^n = \frac{1}{t^{4n}} \sum_{n=0}^{\infty} (-1)^n x^{4n}$$

Thus; the limit may be expressed as:

$$L = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} \frac{1}{t^2} \cdot \frac{(-1)^n}{t^{4n}} \cdot \frac{1}{m!} \int_0^t x^{4n+m} dx \quad (t \rightarrow 0)$$

However; $\int_0^t x^{4n+m} = \left[\frac{x^{4n+m+1}}{4n+m+1} \right]_0^t = \frac{t^{4n+m+1}}{4n+m+1}$

And thus:

$$L = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} \frac{(-1)^n}{m!} \cdot \frac{t^{m-1}}{4n+m+1} \quad (t \rightarrow 0)$$

Note that for $m > 1$; the summand vanishes; and thus the sum can be effectively written as:

$$L = \lim_{t \rightarrow 0} \sum_{n=0}^{\infty} \frac{(-1)^n}{m!} \cdot \frac{t^{m-1}}{4n+m+1} \Bigg|_{m=1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{4n+2}$$

Now, note that:

$$\sum_{k=0}^{\infty} (-1)^k x^{4k} = \sum_{k=0}^{\infty} (-x^4)^k = \frac{1}{1+x^4}$$

$$\Rightarrow \sum_{k=0}^{\infty} (-1)^k x^{4k+1} = \frac{x}{1+x^4}$$

$$\Rightarrow \sum_{k=0}^{\infty} (-1)^k \int_0^1 x^{4k+1} dx = \int_0^1 \frac{x dx}{1+x^4}$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{4k+2} = \int_0^1 \frac{x dx}{1+x^4}$$

Thus;

$$\begin{aligned} \mathcal{L} &= \int_0^1 \frac{x dx}{1+x^4} = \frac{1}{2} \cdot \int_0^1 \frac{dt}{1+t^2} [t=x^2] \\ &= \frac{1}{2} \left[\arctan(t) \right]_0^1 \\ &= \frac{1}{2} \cdot \left[\frac{\pi}{4} - 0 \right] = \frac{\pi}{8} \end{aligned}$$

$$\Rightarrow \boxed{\mathcal{L} = \frac{\pi}{8}}$$

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